and this happens only if these values are all equal. But if the values are all equal, the pairs (R_l, C_j) and (R_i, C_m) must be in equilibrium too for, occurring as they do in the same rows and columns as other equilibrium pairs having their value, they must be the minimal values for their rows and maximal values for their columns.

PROBLEMS

1. Prove that if a game has a dominant row and column, that row and column determine an equilibrium pair for the game.
2. Prove that every strategy pair that passes the equilibrium test is in equilibrium in the originally defined sense, and conversely.
3. Give an example of a game with an equilibrium pair whose row is dominated.
4. Suppose in game G row R is strictly dominated by some row S, that is, every entry in S is greater than its correspondent in R. Can R be half of an equilibrium pair for G?

5-3a. Mixed Strategies

If every zero sum game had at least one pair of strategies in equilibrium, the theory of zero sum games would now be complete. Given any game of that kind, we could apply the maximin test and mechanically solve the game. The bad news is that some zero sum games fail to succumb to this method; the good news is that there still is a complete and mathematically rich account of zero sum games.

Consider the following situation. You are playing scissors, paper, and stone against a perfect player who is also a perfect mind reader. Thus whenever you decide on a strategy she picks one that beats it. If you pick paper, she picks scissors; if you pick stone, she picks paper; if you pick scissors, she picks stone. Look at game table 5-15. No pair of strategies is in equilibrium; so you cannot

5-15	*SC*	*PA*	*ST*
SC	0	1	−1
PA	−1	0	1
ST	1	−1	0

even guarantee yourself a draw.

There is a way out of this trap and it leads to a general solution for games of this type. Suppose that instead of picking your strategy directly you choose it by means of a chance mechanism that can yield any of the three possible strategies. For example, you could roll a die and then play play scissors on 1 or 2, paper on 3 or 4, and stone on 5 or 6. In effect you have adopted a strategy that is a mixture of your original strategies and that, when played against any strategy of your opponent's, yields an equal chance of the outcomes determined by your original strategies. In particular, when played against stone your strat-

egy has a probability of 1/3 of yielding -1, the same probability of yielding 0 and the same probability of yielding 1. Thus its expected utility is 0, which is an improvement over your previous situation with its expected utility of -1. It is also easy to see that this mixed strategy has the same expected utility when played against your opponent's strategy of paper or scissors.

By picking a mixed strategy you also defuse your opponent's power to read your mind. For once you have decided on a mixed strategy, you can turn the rest of the game over to a suitably constructed machine. For example, the machine could consist of a drum spinning inside a box with a window that exposes only a third of the drum's surface at any one time—like the slot machines in Las Vegas. Turning the machine on would give the drum an initial spin. The drum would eventually stop and, with a probability determined by the fraction of the drum allocated to each word, display one of the words "scissors," "paper," or "stone." The word displayed would then be used to compute the outcome of the game against your opponent's choice. Since you would have chosen no specific strategy prior to turning the machine on, and since once the machine is on you have no further choices to make, it would be impossible for your opponent to choose a winning counterstrategy by reading your mind.

By the way, this is not a farfetched idea. In war, where the element of surprise can make all the difference, the threat of espionage is ever present. One way to protect oneself is to develop several strategies and choose to act on one only at the last possible minute. Letting a chance mechanism make the choice for you is a refinement of this idea.

So far we appear to have improved your situation, but can we not expect your opponent to counter with a mixed strategy of her own? Yes, but it makes no difference in this game. Your mixed strategy has the expected utility of 0 against any strategy she plays, mixed or not. Before trying to show this we must introduce some terminology and additional symbolism. Let us call the original strategies *pure strategies* to distinguish them from mixed ones. Both you and your opponent have the same pure strategies in this game, namely, *SC*, *PA*, *ST*. So far we have considered one mixed strategy for you, namely, one that involves an equal chance of playing *SC*, *PA*, or *ST*. We can write this as

$$(1/3\ SC,\ 1/3\ PA,\ 1/3\ ST).$$

There are other mixed strategies available to you; for instance, you could play *SC* with probability of 1/2 and *PA* and *ST* each with a probability of 1/4. We can represent this by

$$(1/2\ SC,\ 1/4\ PA,\ 1/4\ ST).$$

More generally, we can represent any of your mixed strategies by

$$(x\ SC,\ y\ PA,\ z\ ST),$$

where x, y, and z are real numbers between 0 and 1, inclusively, which sum to 1. These numbers give the probability with which one of the pure strategies will be played. In the game at hand both players happen to have three strategies, but in other games they might have a different number of strategies. Suppose the

pure strategies available to Row in a game are $R_1, R_2, \ldots, R_n$. Then his mixed strategies take the form

$$(x_1 R_1, x_2 R_2, \ldots, x_n R_n);$$

where each x_i is a real number between 0 and 1 inclusively, and the sum of the x_i's equals 1. Col's mixed strategies are given a similar representation.

Notice that pure strategies are special cases of mixed ones. The pure strategy *ST* is equivalent to the mixed strategy

(0 *SC*, 0 *PA*, 1 *ST*),

in which the probability of playing *ST* is 1.

Now let us return to our example. You are playing the mixed strategy (1/3 *SC*, 1/3 *PA*, 1/3 *ST*); your opponent is going to play some unknown mixed strategy (x *SC*, y *PA*, z *ST*). (Given what has just been said, this also allows for her playing any of her pure strategies as well.) Let us calculate your expected utility for this pair of strategies. Think of your situation as involving a compound gamble. The larger gamble gives you an equal chance of playing *SC*, *PA* or *ST*. Having played these you engage in one of three smaller gambles determined by the mixed strategy used by your opponent. You have a 1/3 chance of playing *SC*, and if you play it you have an x chance of having that met by *SC*, a y chance of having it met by *PA*, and a z chance of having it met by *ST*. Your payoffs are 0, 1, and -1, respectively, for those combinations of strategies. Thus the expected utility of this smaller gamble is $0x + 1y + -1z$. You also have a 1/3 chance of playing *PA* and *ST* and your expected utilities for the smaller gambles resulting from those strategies are $-1x + 0y + 1z$ and $1x + -1y + 0z$, respectively. Thus your expected utility from the strategy as a whole is

$$\begin{aligned}
&1/3\ (0x + 1y + -1z) + \\
&1/3\ (-1x + 0y + 1z) + \\
&1/3\ (1x + -1y + 0z).
\end{aligned}$$

Since each component has the same factor of 1/3, we can add up all the x components, all the y components, and all the z components without explicitly distributing 1/3 through the entire sum. I have arranged these components in columns so that you can see at a glance that each sums to 0. Thus your expected utility for this strategy must be 0 too. This is exactly the same value we observed when we calculated your expected utility for (1/3 *SC*, 1/3 *PA*, 1/3 *ST*) against your opponent's use of the pure strategy *ST*. (This equals (0 *SC*, 0 *PA*, 1 *ST*); so our first calculation is but a special case of the general one given just now.)

So long as you play (1/3 *SC*, 1/3 *PA*, 1/3 *ST*) you are guaranteed an expected utility of 0, and since we are assuming that you have a utility function satisfying the expected utility condition, this means that the utility of this strategy to you is 0. We have also found your security level for the game. But we have not found an equilibrium pair yet. If your opponent plays the pure strategy of *SC*, then although you can guarantee yourself a utility of 0 by playing the mixed (1/3 *SC*, 1/3 *PA*, 1/3 *ST*), you can do better against her *SC*—you can play *ST* and win. She cannot do better, of course, but since you can, the strate-

gies are not in equilibrium. On the other hand, suppose your opponent plays the same mixed strategy you have been playing. Then she can hold your winnings to 0. So the pair of mixed strategies are in equilibrium, and for this particular game they are the only such pair. The game is thus fully solved; we have found its equilibrium pairs and its value.

Before proceeding further you should try to verify that the pair of mixed strategies that are in equilibrium in tables 5-16 and 5-17 are (1/2 R_1, 1/2 R_2) and (1/2 C_1, 1/2 C_2). What are the values of the games?

5-16

	C_1	C_2
R_1	1	−1
R_2	−1	1

5-17

	C_1	C_2
R_1	22	−18
R_2	−18	22

All the games we have considered so far have been symmetrical games and it has been easy to guess their equilibrium pairs. Finding the equilibrium pairs for the next game (table 5-18) is much harder. If we try the mixed strategy (1/2 R_1, 1/2 R_2) against C_1, we find that its expected utility is 4 whereas its expected utility against C_2 is 7/2. Furthermore, the mixed strategy (1/2 C_1, 1/2 C_2) for Col has an expected utility of 9/2 against R_1 and 3 against R_2.

5-18

	C_1	C_2
R_1	6	3
R_2	2	4

The joint expected utility of the two mixed strategies is

$$(1/2)(1/2)6 + (1/2)(1/2)3 + (1/2)(1/2)2 + (1/2)(1/2)4 = 3\ 3/4.$$

Thus they cannot be an equilibrium pair, since Row is better off playing R_1 when Col plays his half. We can keep testing pairs of strategies to see if they are in equilibrium, but there are infinitely many to test and no guarantee that our best guesses will find one that works. Plainly, a systematic approach is required.

To this end, let p be the probability of playing R_1 for any given mixed strategy of Row's and let q be the probability of playing C_1 for any given mixed strategy of Col's. Then the expected utility for Row of these strategies being played jointly is given by

$$EU(p, q) = 6pq + 3p(1-q) + 2(1-p)q + 4(1-p)(1-q),$$

since there is a pq chance of ending with a 6, a $(1-p)q$ chance of ending with 3, and so on. Applying some algebra we can convert this equation to

$$EU(p, q) = 5pq - p - 2q + 4.$$

Now Row's object is to choose p so as to maximize this number while Col's is

to choose q so as to minimize it. So let us do some fancy factoring to isolate Row's and Col's contributions to this number. We proceed as follows:

$$\begin{aligned} EU(p, q) &= 5(pq - p/5 - 2q/5 + 4/5) \\ &= 5(pq - p/5 - 2q/5 + 2/25 + 18/25) \\ &= 5(pq - p/5 - 2q/5 + 2/25) + 18/5 \\ &= 5[(p - 2/5)(q - 1/5)] + 18/5. \end{aligned}$$

If Row picks p as 2/5, his factor equals 0 and the value of $EU(p, q)$ is 18/5 no matter what q is; that is, $EU(2/5, q) = 18/5$ for all q. Similarly, $EU(p, 1/5) = 18/5$ for all p. Thus although each player might do better with a different p or q provided the other player played a bad strategy, we have found the security strategies for both players and they are in equilibrium. Thus we have obtained at least one equilibrium pair for this game.

Might there be more than one equilibrium pair? Suppose p_1 and q_1 also determine an equilibrium pair; that is, $(p_1 R_1, (1 - p_1) R_2)$ and $(q_1 C_1, (1 - q_1) C_2)$ are also in equilibrium. Then $EU(p_1, q_1) \leq EU(p_1, q)$ for all choices of q; otherwise Col would want to change when Row played the strategy determined by p_1. Similarly, $EU(p, q_1) \leq EU(p_1, q_1)$ for all choices of p. But we already know that

$$EU(p_1, 1/5) = 18/5 = EU(2/5, q_1),$$

and since $EU(p_1, q_1)$ is bounded above and below by those two identical quantities, it must equal them. In other words, if there are two equilibrium pairs, they give rise to the same security levels. It is also easily shown that if p_1 and q_1 determine an equilibrium pair, so do 2/5 and q_1 and p_1 and 1/5. We thus get a mixed-strategy version of the coordination theorem for this game.

Every two-person zero sum game has a solution. This follows immediately from the next theorem.

THE MAXIMIN THEOREM FOR TWO-PERSON ZERO SUM GAMES: *For every two-person zero sum game there is at least one strategy (mixed or pure) for Row and at least one strategy for Col that form an equilibrium pair. If there is more than one such pair, their expected utilities are equal.*

The expected utility for the equilibrium pair is called *the value* v of the game. It is the security level for the game for both Row and Col. In other words, by playing his half of the equilibrium pair Row can be sure of winning at least v, and by playing his half Col can be sure of holding Row's winnings to no more than v. Row's equilibrium strategy is called a *maximin strategy* because it maximizes his minimal outcomes; Col's is called a *minimax* strategy because it holds Row's maximums to a minimum. The maximin theorem is commonly approached from Col's more defensive viewpoint and called "the minimax theorem" instead. I am using the other name because my exposition has emphasized Row's point of view.

There are a number of proofs of the maximin theorem in the literature, but they all involve mathematical techniques far beyond those presupposed by this book. However, the method used in the last example can be generalized to

yield a simple proof of the maximin theorem for two-by-two games. So that will be our next topic.

PROBLEMS

1. Suppose Col has k pure strategies. What is the general form of his mixed strategies?
2. Show that in any game with the form shown in table 5-19 the strategies

5-19

a	$-b$
$-b$	a

$(1/2\ R_1,\ 1/2\ R_2)$ and $(1/2\ C_1,\ 1/2\ C_2)$ are in equilibrium.

3. State precisely and prove the mixed-strategy coordination theorem for two-by-two zero sum games.

5-3b. Proof of the Maximin Theorem for Two-by-Two Games

In order to simplify our proof we will consider only games in a restricted form. It will be called the standard form (SF) and is represented by table 5-20. Here

5-20

(SF)

	C_1	C_2
R_1	a	b
R_2	c	d

a, b, c, and d must all be positive numbers and

$$a > c,\ d > b, \text{ and } d > c.$$

Restricting ourselves to games in standard form will not limit the scope of our theorem, because, as the exercises in the next Problems section show, any two-by-two game that is not in SF is either solvable by our previous methods or is equivalent to a game in SF from which its solution is easily obtained.

We can assume, then, that all the games under consideration are in SF. As we have already observed in dealing with preceding examples, the expected utility for any pair of strategies

$$(p\ R_1,\ (1-p)\ R_2);\ (q\ C_1,\ (1-q)\ C_2)$$

is given by the formula

$$EU(p,\ q) = apq + bp(1-q) + c(1-p)q + d(1-p)(1-q).$$

Applying some algebra to this, we obtain

$$\begin{aligned} EU(p,\ q) &= apq + bp - bpq + cq - cpq + d - dp - dq + dpq \\ &= (a - c + d - b)pq + (b - d)p + (c - d)q + d. \end{aligned}$$

Given that $a > c$ and $d > b$, the sum $(a - c + d - b)$ must be positive while